

Chapter 2 Assignment Due date: March 30, 2026 (Mon)

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2.1

Sol.

i.

$$f(x) = F'(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 1, & \text{o.w.} \end{cases}$$

ii.

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{a+b}{2} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x^2 dx \\ &= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3} \end{aligned}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 = \frac{(b-a)^2}{12}$$

- iii.
- The pdf is constant on $[a, b]$, so all values in the interval are equally likely.
 - The graph of pdf is a rectangle over the interval $[a, b]$.

iv.

$$P(7 \leq X \leq 9) = \frac{9-7}{10-5} = \frac{2}{5} = 0.4$$

2.6

Sol.

$$X \sim \text{Bin}(n = 10, p = 0.2)$$

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

i.

$$E(X) = np = 10 \cdot 0.2 = 2$$

$$\text{Var}(X) = np(1-p) = 2 \cdot (1-0.2) = 1.6$$

ii.

$$\begin{aligned} P(3 \leq X \leq 5) &= P(X=3) + P(X=4) + P(X=5) \\ &= \binom{10}{3} (0.2)^3 (0.8)^7 + \binom{10}{4} (0.2)^4 (0.8)^6 + \binom{10}{5} (0.2)^5 (0.8)^5 \\ &\approx 0.3158 \end{aligned}$$

2.8

Sol.

10, 12, 15, 17, 20, 20, 23, 25, 28, 30, 40, 43, 45, 48, 50, 50, 59, 61, 64, 70.

$$\text{i. } 20 \cdot \frac{1}{4} = 5 \Rightarrow Q_1 = \frac{a_5 + a_6}{2} = \frac{20 + 20}{2} = 20$$

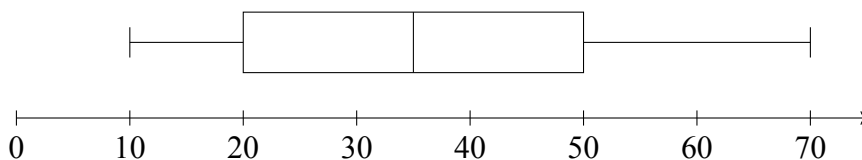
$$20 \cdot \frac{1}{2} = 10 \Rightarrow Me = \frac{a_1 0 + a_1 1}{2} = \frac{30 + 40}{2} = 35$$

$$20 \cdot \frac{3}{4} = 15 \Rightarrow Q_3 = \frac{a_1 5 + a_1 6}{2} = \frac{50 + 50}{2} = 50$$

$$\text{ii. } IQR = Q_3 - Q_1 = 50 - 20 = 30$$

$$\text{iii. Lower bound} = Q_1 - 1.5IQR = 20 - 45 = -25$$

$$\text{Upper bound} = Q_3 + 1.5IQR = 50 + 45 = 95$$



2.9

Sol.

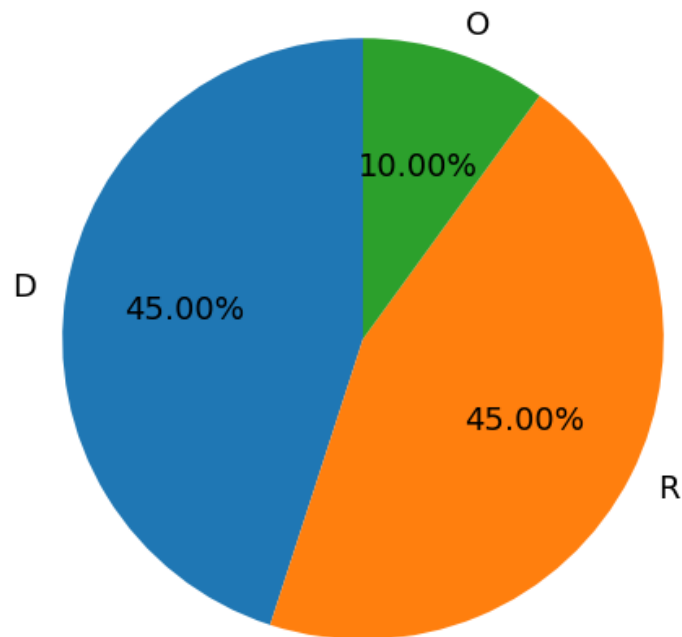
D, D, R, O, R, R, R, D, R, D, D, R, D, D, R, R, R, O, D, D

i.

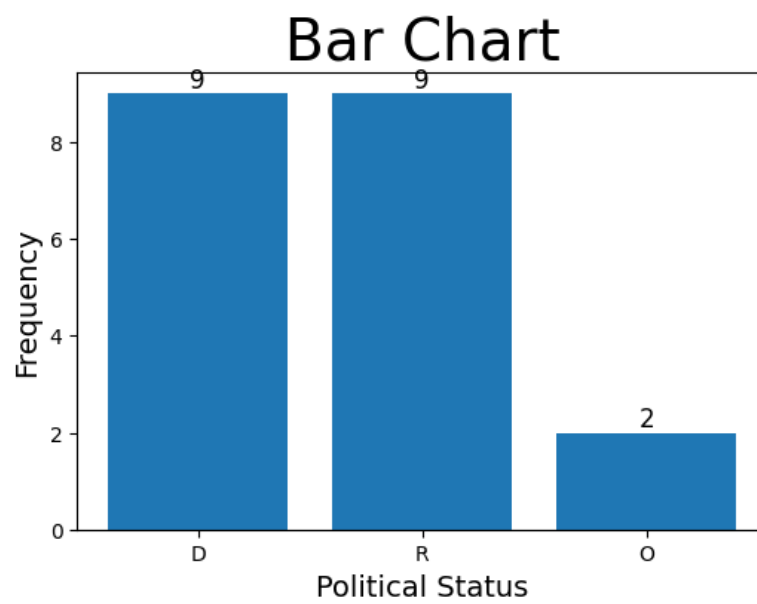
Political Status	Frequency	Relative Frequency
<i>D</i> (Democrats)	9	0.45
<i>R</i> (Republicans)	9	0.45
<i>O</i> (Others)	2	0.10

ii.

Pie Chart



iii.



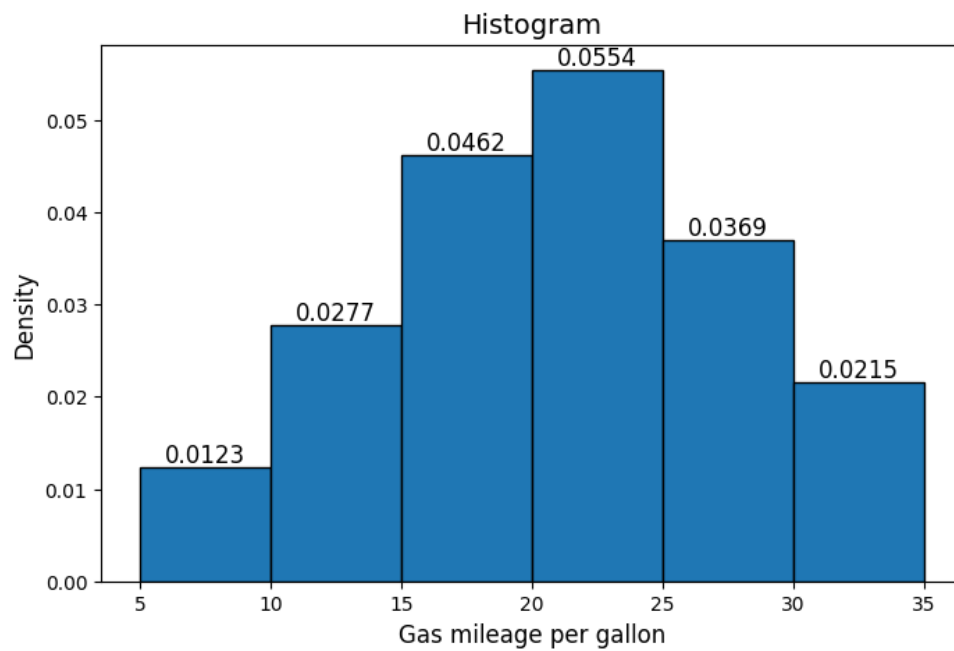
2.11

Sol.

i.

Class Interval	Frequency	Relative Frequency	Density
[5, 10)	20	0.0615	0.0123
[10, 15)	45	0.1385	0.0277
[15, 20)	75	0.2308	0.0462
[20, 25)	90	0.2769	0.0554
[25, 30)	60	0.1846	0.0369
[30, 35)	35	0.1077	0.0215

ii.



iii.

$$P(5 \leq X < 20) = \frac{20 + 45 + 75}{325} \approx 0.4308$$

2.14

Sol.

2.19

Sol.

2.21

Sol.

2.25

Sol.

2.26

Sol.